

Final Project

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Read in the cogimp0 dataset

```
# Access and read the cogimp0 file from the drive
cogimp0 = drive_download(
  as_id("1xDBY_8-djIWfKWtBKrx1YbZ9mkw6z1J2"),
  path = tempfile(fileext = ".sav"),
  overwrite = TRUE
) %>%
  { read_sav(.$local_path) }

## Auto-refreshing stale OAuth token.
## File downloaded:
## * 'COGIMP9220A_R.sav' <id: 1xDBY_8-djIWfKWtBKrx1YbZ9mkw6z1J2>
## Saved locally as:
## * '/var/folders/_d/fx514zw972j96qby8l86rsgh0000gn/T//RtmpqSj8Rp/file84061cb7c6be.sav'
# generate codebook dataframe
cogimp0_codebook = labelled::generate_dictionary(cogimp0)
```

Read in the 2022 tracker file dataset

```
# read in the trk2022v2 file
trk2022v2 = drive_download(
  as_id("1G0wZriMuTMftZYP834aJG7nMrN3k4T5E"),
  path = tempfile(fileext = ".sav"),
  overwrite = TRUE
) %>%
  { read_sav(.$local_path) }

## File downloaded:
## * 'trk2022tr_r.sav' <id: 1G0wZriMuTMftZYP834aJG7nMrN3k4T5E>
## Saved locally as:
## * '/var/folders/_d/fx514zw972j96qby8l86rsgh0000gn/T//RtmpqSj8Rp/file840638d429a2.sav'
# generate codebook
trk2022v2_codebook = labelled::generate_dictionary(trk2022v2)
```

Combine the data and variables we need

```
cogimp0 <- cogimp0 %>% mutate(hhid = as.character(HHID), pn = as.character(PN))
trk2022v2 <- trk2022v2 %>% mutate(hhid = as.character(hhid), pn = as.character(pn))

data_w13 <- cogimp0 %>%
  dplyr::select(hhid, pn, R13M0, R13DY, R13YR, R13DW, R13SCIS, R13CACT, R13PRES, R13VP) %>% rename(
    Month= R13M0,
    Day= R13DY,
    Year= R13YR,
    DOW= R13DW,
    Scissors= R13SCIS,
    Cactus= R13CACT,
    President= R13PRES,
    VP = R13VP
  ) %>%
  mutate(wave = "W13")

data_w3 <- cogimp0 %>%
  dplyr::select(hhid, pn, R3M0, R3DY, R3YR, R3DW, R3SCIS, R3CACT, R3PRES, R3VP) %>%
  rename(
    Month= R3M0,
    Day= R3DY,
    Year= R3YR,
    DOW = R3DW,
    Scissors = R3SCIS,
    Cactus = R3CACT,
    President = R3PRES,
    VP = R3VP
  ) %>%
  mutate(wave = "W3")

# data_long <- bind_rows(data_w3, data_w13)

cov_w13 <- trk2022v2 %>%
  dplyr::select(hhid, pn, page, gender, degree, race, immgyear, usborn, hispanic) %>%
  rename(age = page) %>%
  mutate(wave = "W13")

cov_w3 <- trk2022v2 %>%
  dplyr::select(hhid, pn, eage, gender, degree, race, immgyear, usborn, hispanic) %>%
  rename(age = eage) %>%
  mutate(wave = "W3")

# covs_long <- bind_rows(cov_w3, cov_w13)

# lca_raw <- data_long %>%
#   inner_join(covs_long, by = c("hhid", "pn", "wave"))

# indicator_names <- c("Month", "Day", "Year", "DOW", "Scissors", "Cactus", "President", "VP")

lca_w3 <- data_w3 %>%
  inner_join(cov_w3, by = c("hhid", "pn", "wave"))

indicator_names <- c("Month", "Day", "Year", "DOW", "Scissors", "Cactus", "President", "VP")
```

```
#lca_ready <- lca_raw %>%
# dplyr::mutate(across(all_of(indicator_names),
# ~ ifelse(is.na(.), NA, as.integer(. + 1))))
```

```
lca_ready_w3 <- lca_w3 %>%
  dplyr::mutate(across(all_of(indicator_names),
    ~ ifelse(is.na(.), NA, as.integer(. + 1))))
```

```
summary(lca_w3[, indicator_names])
```

```
##      Month      Day      Year      DOW
## Min.   :0.000 Min.   :0.000 Min.   :0.000 Min.   :0.000
## 1st Qu.:1.000 1st Qu.:1.000 1st Qu.:1.000 1st Qu.:1.000
## Median :1.000 Median :1.000 Median :1.000 Median :1.000
## Mean   :0.974 Mean   :0.835 Mean   :0.975 Mean   :0.976
## 3rd Qu.:1.000 3rd Qu.:1.000 3rd Qu.:1.000 3rd Qu.:1.000
## Max.   :1.000 Max.   :1.000 Max.   :1.000 Max.   :1.000
## NA's   :24690 NA's   :24690 NA's   :24690 NA's   :24690
##      Scissors      Cactus      President      VP
## Min.   :0.000 Min.   :0.000 Min.   :0.000 Min.   :0.000
## 1st Qu.:1.000 1st Qu.:1.000 1st Qu.:1.000 1st Qu.:0.000
## Median :1.000 Median :1.000 Median :1.000 Median :1.000
## Mean   :0.988 Mean   :0.894 Mean   :0.938 Mean   :0.711
## 3rd Qu.:1.000 3rd Qu.:1.000 3rd Qu.:1.000 3rd Qu.:1.000
## Max.   :1.000 Max.   :1.000 Max.   :1.000 Max.   :1.000
## NA's   :24690 NA's   :24690 NA's   :24690 NA's   :24690
```

```
summary(lca_w3$age)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      21.0   67.0   999.0   751.3   999.0   999.0
```

```
table(lca_w3$gender, useNA = "ifany")
```

```
##
##      1      2
## 17830 23211
```

```
table(lca_w3$degree, useNA = "ifany")
```

```
##
##      0      1      2      3      4      5      6      9
## 10117 2015 17600 2041 4771 2418 797 1282
```

```
table(lca_w3$race, useNA = "ifany")
```

```
##
##      0      1      2      7
## 111 29627 7871 3432
```

```
table(lca_w3$wave)
```

```
##
##      W3
## 41041
```

Fit LCA model

Without covariates

```
set.seed(687)
f_ind <- cbind(Month, Day, Year, DOW, Scissors, Cactus, President, VP) ~ 1
sub_df <- lca_ready_w3 %>% dplyr::sample_frac(0.5)
fit_list <- lapply(2:5, function(k) {
  polCA(f_ind, data = sub_df, nclass = k, maxiter = 1500, nrep = 3, verbose = FALSE)
})

data.frame(
  k = 2:5,
  AIC = sapply(fit_list, \(m) m$aic),
  BIC = sapply(fit_list, \(m) m$bic),
  Gsq = sapply(fit_list, \(m) m$Gsq)
)
```

```
##   k      AIC      BIC      Gsq
## 1 2 30771.76 30890.96 683.0069
## 2 3 30435.14 30617.44 328.3878
## 3 4 30328.27 30573.67 203.5165
## 4 5 30309.01 30617.52 166.2595
```

We will use 4 classes with BIC the lowest for w3.

```
set.seed(687)
lca_k4_w3 <- polCA(f_ind, data = lca_ready_w3, nclass = 4, maxiter = 5000, nrep = 20, verbose = TRUE)

## Model 1: llik = -30487.05 ... best llik = -30487.05
## Model 2: llik = -30491.71 ... best llik = -30487.05
## Model 3: llik = -30487.05 ... best llik = -30487.05
## Model 4: llik = -30487.05 ... best llik = -30487.05
## Model 5: llik = -30487.05 ... best llik = -30487.05
## Model 6: llik = -30487.05 ... best llik = -30487.05
## Model 7: llik = -30491.71 ... best llik = -30487.05
## Model 8: llik = -30487.05 ... best llik = -30487.05
## Model 9: llik = -30487.05 ... best llik = -30487.05
## Model 10: llik = -30487.05 ... best llik = -30487.05
## Model 11: llik = -30487.05 ... best llik = -30487.05
## Model 12: llik = -30487.05 ... best llik = -30487.05
## Model 13: llik = -30487.05 ... best llik = -30487.05
## Model 14: llik = -30487.05 ... best llik = -30487.05
## Model 15: llik = -30487.05 ... best llik = -30487.05
## Model 16: llik = -30487.05 ... best llik = -30487.05
## Model 17: llik = -30487.05 ... best llik = -30487.05
## Model 18: llik = -30487.05 ... best llik = -30487.05
## Model 19: llik = -30491.71 ... best llik = -30487.05
## Model 20: llik = -30487.05 ... best llik = -30487.05
## Conditional item response (column) probabilities,
## by outcome variable, for each class (row)
##
## $Month
##           Pr(1) Pr(2)
## class 1:  0.6334 0.3666
## class 2:  0.0055 0.9945
```

```

## class 3: 0.0292 0.9708
## class 4: 0.1958 0.8042
##
## $Day
##           Pr(1) Pr(2)
## class 1: 0.9003 0.0997
## class 2: 0.1058 0.8942
## class 3: 0.2800 0.7200
## class 4: 0.6655 0.3345
##
## $Year
##           Pr(1) Pr(2)
## class 1: 0.8287 0.1713
## class 2: 0.0019 0.9981
## class 3: 0.0369 0.9631
## class 4: 0.1428 0.8572
##
## $DOW
##           Pr(1) Pr(2)
## class 1: 0.4647 0.5353
## class 2: 0.0048 0.9952
## class 3: 0.0312 0.9688
## class 4: 0.2186 0.7814
##
## $Scissors
##           Pr(1) Pr(2)
## class 1: 0.0865 0.9135
## class 2: 0.0065 0.9935
## class 3: 0.0264 0.9736
## class 4: 0.0354 0.9646
##
## $Cactus
##           Pr(1) Pr(2)
## class 1: 0.6211 0.3789
## class 2: 0.0491 0.9509
## class 3: 0.3399 0.6601
## class 4: 0.1829 0.8171
##
## $President
##           Pr(1) Pr(2)
## class 1: 0.7079 0.2921
## class 2: 0.0109 0.9891
## class 3: 0.2893 0.7107
## class 4: 0.0000 1.0000
##
## $VP
##           Pr(1) Pr(2)
## class 1: 0.9539 0.0461
## class 2: 0.1626 0.8374
## class 3: 0.8974 0.1026
## class 4: 0.2877 0.7123
##
## Estimated class population shares
## 0.0147 0.7973 0.1494 0.0385

```

```
##
## Predicted class memberships (by modal posterior prob.)
## 0.0143 0.8428 0.126 0.0169
##
## =====
## Fit for 4 latent classes:
## =====
## number of observations: 16351
## number of estimated parameters: 35
## residual degrees of freedom: 220
## maximum log-likelihood: -30487.05
##
## AIC(4): 61044.1
## BIC(4): 61313.67
## G^2(4): 289.0484 (Likelihood ratio/deviance statistic)
## X^2(4): 446.6359 (Chi-square goodness of fit)
##
```

```
lca_k4_w3$probs
```

```
## $Month
##           Pr(1)      Pr(2)
## class 1: 0.633396914 0.3666031
## class 2: 0.005508788 0.9944912
## class 3: 0.029248199 0.9707518
## class 4: 0.195826123 0.8041739
##
## $Day
##           Pr(1)      Pr(2)
## class 1: 0.9003198 0.09968016
## class 2: 0.1057820 0.89421798
## class 3: 0.2799834 0.72001660
## class 4: 0.6654896 0.33451038
##
## $Year
##           Pr(1)      Pr(2)
## class 1: 0.828735546 0.1712645
## class 2: 0.001891506 0.9981085
## class 3: 0.036864284 0.9631357
## class 4: 0.142788567 0.8572114
##
## $DOW
##           Pr(1)      Pr(2)
## class 1: 0.464695278 0.5353047
## class 2: 0.004776264 0.9952237
## class 3: 0.031208886 0.9687911
## class 4: 0.218567216 0.7814328
##
## $Scissors
##           Pr(1)      Pr(2)
## class 1: 0.086456894 0.9135431
## class 2: 0.006483089 0.9935169
## class 3: 0.026353363 0.9736466
## class 4: 0.035395135 0.9646049
##
```

```
## $Cactus
##           Pr(1)      Pr(2)
## class 1: 0.62110589 0.3788941
## class 2: 0.04907937 0.9509206
## class 3: 0.33994877 0.6600512
## class 4: 0.18292733 0.8170727
##
## $President
##           Pr(1)      Pr(2)
## class 1: 7.079249e-01 0.2920751
## class 2: 1.089080e-02 0.9891092
## class 3: 2.893035e-01 0.7106965
## class 4: 3.740145e-12 1.0000000
##
## $VP
##           Pr(1)      Pr(2)
## class 1: 0.9538861 0.04611392
## class 2: 0.1626035 0.83739647
## class 3: 0.8974073 0.10259266
## class 4: 0.2877396 0.71226043
```

```
lca_ready_w3$rowid <- seq_len(nrow(lca_ready_w3))
mf <- model.frame(f_ind, data = lca_ready_w3, na.action = na.omit)
idx <- as.integer(row.names(mf))
tab_w3 <- table(lca_k4_w3$predclass, lca_ready_w3$wave[idx])
tab_w3
```

```
##
##      W3
##  1   234
##  2 13780
##  3  2060
##  4   277
```

With covariates

```
set.seed(687)
f_cov <- cbind(Month, Day, Year, DOW, Scissors, Cactus, President, VP) ~
  age + factor(gender) + factor(degree) + factor(race) + factor(usborn) + factor(hispanic)

lca_cov_w3 <- poLCA(f_cov, data = lca_ready_w3, nclass = 4, maxiter = 5000, nrep = 20, verbose = TRUE)
```

```
## Model 1: llik = -29962.38 ... best llik = -29962.38
## Model 2: llik = -29299.16 ... best llik = -29299.16
## Model 3: llik = -29631.56 ... best llik = -29299.16
## Model 4: llik = -29167.72 ... best llik = -29167.72
## Model 5: llik = -33222.14 ... best llik = -29167.72
## Model 6: llik = -29023.88 ... best llik = -29023.88
## Model 7: llik = -29329.09 ... best llik = -29023.88
## Model 8: llik = -30481.26 ... best llik = -29023.88
## Model 9: llik = -29056.15 ... best llik = -29023.88
## Model 10: llik = -30023.66 ... best llik = -29023.88
## Model 11: llik = -32462.88 ... best llik = -29023.88
## Model 12: llik = -29624.76 ... best llik = -29023.88
## Model 13: llik = -30173.38 ... best llik = -29023.88
## Model 14: llik = -29307.34 ... best llik = -29023.88
```

```

## Model 15: llik = -30740.11 ... best llik = -29023.88
## Model 16: llik = -29497.52 ... best llik = -29023.88
## Model 17: llik = -30517.08 ... best llik = -29023.88
## Model 18: llik = -29526.08 ... best llik = -29023.88
## Model 19: llik = -29688.87 ... best llik = -29023.88
## Model 20: llik = -32412.2 ... best llik = -29023.88
## Conditional item response (column) probabilities,
## by outcome variable, for each class (row)
##
## $Month
##           Pr(1) Pr(2)
## class 1: 0.0403 0.9597
## class 2: 0.0098 0.9902
## class 3: 0.4598 0.5402
## class 4: 0.0238 0.9762
##
## $Day
##           Pr(1) Pr(2)
## class 1: 0.2076 0.7924
## class 2: 0.1216 0.8784
## class 3: 0.8266 0.1734
## class 4: 0.2365 0.7635
##
## $Year
##           Pr(1) Pr(2)
## class 1: 0.0347 0.9653
## class 2: 0.0037 0.9963
## class 3: 0.6230 0.3770
## class 4: 0.0205 0.9795
##
## $DOW
##           Pr(1) Pr(2)
## class 1: 0.0038 0.9962
## class 2: 0.0103 0.9897
## class 3: 0.4030 0.5970
## class 4: 0.0249 0.9751
##
## $Scissors
##           Pr(1) Pr(2)
## class 1: 0.0111 0.9889
## class 2: 0.0060 0.9940
## class 3: 0.0833 0.9167
## class 4: 0.0244 0.9756
##
## $Cactus
##           Pr(1) Pr(2)
## class 1: 0.0368 0.9632
## class 2: 0.0169 0.9831
## class 3: 0.5076 0.4924
## class 4: 0.4096 0.5904
##
## $President
##           Pr(1) Pr(2)
## class 1: 0.1688 0.8312

```



```

## class 2: 0.0139 0.9861
## class 3: 0.5382 0.4618
## class 4: 0.1721 0.8279
##
## $VP
##           Pr(1) Pr(2)
## class 1: 0.8980 0.1020
## class 2: 0.1527 0.8473
## class 3: 0.8507 0.1493
## class 4: 0.6701 0.3299
##
## Estimated class population shares
## 0.0239 0.757 0.0275 0.1916
##
## Predicted class memberships (by modal posterior prob.)
## 0.0191 0.787 0.0248 0.169
##
## =====
## Fit for 4 latent classes:
## =====
## 2 / 1
##           Coefficient Std. error t value Pr(>|t|)
## (Intercept)      0.98152    0.07594   12.924   0.000
## age             -0.00143    0.00029   -5.011   0.000
## factor(gender)2    0.80773    0.06279   12.865   0.000
## factor(degree)1    0.97872    0.13422    7.292   0.000
## factor(degree)2    2.91735    0.06061   48.130   0.000
## factor(degree)3    2.86297    0.03406   84.052   0.000
## factor(degree)4    1.35847    0.13928    9.753   0.000
## factor(degree)5    2.63327    0.17587   14.973   0.000
## factor(degree)6    2.28381    0.20471   11.156   0.000
## factor(degree)9    0.00576    0.00110    5.219   0.000
## factor(race)1      0.99594    0.06877   14.482   0.000
## factor(race)2      0.07129    0.06435    1.108   0.270
## factor(race)7     -0.09332    0.10642   -0.877   0.382
## factor(usborn)5   -1.99381    0.09733  -20.485   0.000
## factor(usborn)9   -0.00031    0.00180   -0.171   0.864
## factor(hispanic)1 -1.74343    0.10349  -16.847   0.000
## factor(hispanic)2  0.30266    0.12706    2.382   0.018
## factor(hispanic)3  0.17353    0.01017   17.058   0.000
## factor(hispanic)5  2.24358    0.06501   34.513   0.000
## =====
## 3 / 1
##           Coefficient Std. error t value Pr(>|t|)
## (Intercept)     -0.72362    0.10713   -6.755   0.000
## age              0.00053    0.00031    1.717   0.088
## factor(gender)2    0.79707    0.09350    8.525   0.000
## factor(degree)1   -0.44415    0.24379   -1.822   0.070
## factor(degree)2    0.92206    0.10177    9.060   0.000
## factor(degree)3   -2.52714    0.01149  -219.913   0.000
## factor(degree)4   -0.34982    0.09317   -3.755   0.000
## factor(degree)5   -2.44207    0.04507  -54.180   0.000
## factor(degree)6   -2.25071    0.00215 -1049.060   0.000
## factor(degree)9   -0.01085    0.00017  -62.672   0.000

```

```

## factor(race)1      -0.58123    0.08885    -6.542    0.000
## factor(race)2      0.69041    0.08797     7.848    0.000
## factor(race)7     -0.80599    0.14904    -5.408    0.000
## factor(usborn)5    -0.94990    0.14371    -6.610    0.000
## factor(usborn)9    -0.04634    0.00651    -7.114    0.000
## factor(hispanic)1  -0.83667    0.14753    -5.671    0.000
## factor(hispanic)2  -1.44964    0.17473    -8.297    0.000
## factor(hispanic)3  -0.11784    0.00989   -11.915    0.000
## factor(hispanic)5   1.72013    0.10237    16.803    0.000
## =====
## 4 / 1
##               Coefficient Std. error t value Pr(>|t|)
## (Intercept)      1.19854    0.08522   14.065   0.000
## age             -0.00076    0.00030   -2.524   0.013
## factor(gender)2   1.21671    0.07111   17.111   0.000
## factor(degree)1  -0.53313    0.17708   -3.011   0.003
## factor(degree)2   1.09853    0.07463   14.719   0.000
## factor(degree)3  -0.35307    0.02345  -15.055   0.000
## factor(degree)4  -2.32675    0.06473  -35.944   0.000
## factor(degree)5  -0.31311    0.13172   -2.377   0.019
## factor(degree)6   0.02689    0.20178    0.133   0.894
## factor(degree)9   0.01070    0.00072   14.788   0.000
## factor(race)1     -0.15096    0.07380   -2.046   0.042
## factor(race)2      1.36385    0.06235   21.876   0.000
## factor(race)7     -0.01634    0.11206   -0.146   0.884
## factor(usborn)5   -0.93117    0.10724   -8.683   0.000
## factor(usborn)9    0.03434    0.00529    6.497   0.000
## factor(hispanic)1 -1.40985    0.12040  -11.710   0.000
## factor(hispanic)2  0.86870    0.13298    6.532   0.000
## factor(hispanic)3  0.01320    0.00476    2.775   0.006
## factor(hispanic)5  1.71844    0.07081   24.270   0.000
## =====
## number of observations: 16351
## number of estimated parameters: 89
## residual degrees of freedom: 166
## maximum log-likelihood: -29023.88
##
## AIC(4): 58225.76
## BIC(4): 58911.24
## X^2(4): 845.4265 (Chi-square goodness of fit)
##
## ALERT: estimation algorithm automatically restarted with new initial values
##

```

```
lca_cov_w3$P
```

```
## [1] 0.02385792 0.75699372 0.02754029 0.19160807
```

```
lca_cov_w3$coeff
```

```

##               [,1]      [,2]      [,3]
## (Intercept)  0.9815236110 -0.7236228364  1.1985445824
## age         -0.0014315012  0.0005313769 -0.0007589214
## factor(gender)2 0.8077256954 0.7970673262  1.2167073328
## factor(degree)1 0.9787203159 -0.4441534759 -0.5331295437

```

```
## factor(degree)2      2.9173487920  0.9220569097  1.0985276858
## factor(degree)3      2.8629663144 -2.5271436058 -0.3530719937
## factor(degree)4      1.3584654802 -0.3498166864 -2.3267509033
## factor(degree)5      2.6332695825 -2.4420742573 -0.3131063297
## factor(degree)6      2.2838056937 -2.2507122354  0.0268857669
## factor(degree)9      0.0057578004 -0.0108508164  0.0106959353
## factor(race)1        0.9959389532 -0.5812313459 -0.1509554774
## factor(race)2        0.0712908861  0.6904060912  1.3638514984
## factor(race)7       -0.0933196248 -0.8059918239 -0.0163351193
## factor(usborn)5     -1.9938112084 -0.9499049047 -0.9311669447
## factor(usborn)9     -0.0003085191 -0.0463364801  0.0343398354
## factor(hispanic)1  -1.7434272576 -0.8366692916 -1.4098535657
## factor(hispanic)2   0.3026604724 -1.4496418050  0.8686982135
## factor(hispanic)3   0.1735255648 -0.1178445205  0.0132015477
## factor(hispanic)5   2.2435780898  1.7201285090  1.7184404053

# Without covariates
saveRDS(lca_k4_w3, "lca_w3_k4_nocov.rds")
# With covariates
saveRDS(lca_cov_w3, "lca_w3_k4_cov.rds")
```